



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

DISTRIBUTED DATABASE MANAGEMENT SYSTEM [IT-3222]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) You are tasked with designing a distributed database for a global application with a high volume of user data. Analyze using any 2 sharding methods based on the following points:
 - A)
 1. Trade-offs
 2. Challenges
 3. Solutions

Provide your analysis by comparing the approaches, highlighting which method or combination of methods might best suit the application's needs and why.
 - B) Imagine you are designing a distributed DBMS where each site operates autonomously. Apply your understanding of autonomy to explain how this feature impacts performance of the database. Provide a specific example or scenario to illustrate your answer.
 - C) You are designing a banking application that needs to store customer data across different geographic locations. Apply the concept of distributed databases and distributed DBMS to explain how your design ensures data transparency and high availability.
- 2) A multinational e-commerce company, GlobalShop, is planning to implement a homogeneous distributed database system to manage its inventory and sales data across its operations in North America, Europe, and Asia. The company wants to ensure real-time data consistency, efficient management, and uniform access to data across all regions.
 - A) Questions:
 1. Justify the suitability of a homogeneous database for GlobalShop's requirements.
 2. Propose a suitable DBMS for GlobalShop's homogeneous database and justify your choice.
 3. Outline two potential challenges GlobalShop might face when implementing this homogeneous distributed database system and suggest solutions.
 - B) Imagine you are designing a distributed database for a global research collaboration platform. Apply the different orthogonal dimensions to explain the impact of these factors on your design choices to optimize system performance and user concurrency.
 - C) "Referential integrity and domain integrity help maintain data consistency in a

Relational Database Management System (RDBMS)". Elaborate the above statement with an example.

- 3) You are building a file-sharing application using peer-to-peer (P2P) architecture. Apply the principles of P2P data management to explain how you would handle data lookup, availability, and scalability without using a centralized index. (5)
- A)
- B) You are developing a peer-to-peer (P2P) system for collaborative scientific research, where datasets are distributed across multiple peers using different schemas. Apply the reference architecture of P2P data management to explain how your system will handle: (3)
1. A user-submitted global query,
 2. Schema differences between peers, and
 3. Efficient response collection and caching of results.
- C) The web graph is known to be sparse, self-organizing, and a small-world network. Explain with an example the influence of these characteristics for designing of an efficient web search algorithm. (2)
- 4) In a distributed database system, three relations - EMP, ASG, and PROJ - are located on different sites. A query requires joining all three relations:
SELECT ENAME, PNAME
FROM EMP, ASG, PROJ
WHERE EMP.ENO = ASG.ENO AND ASG.PNO = PROJ.PNO;
The sizes of the relations are as follows: (5)
- size(EMP) = 10,000 records
 - size(ASG) = 2,000 records
 - size(PROJ) = 500 records
- The system aims to minimize communication cost while executing the query efficiently.
Analyze the given scenario and recommend a join order that minimizes communication cost. Justify your recommendation based on relation sizes and distributed join characteristics. Additionally, explain whether semi-join strategies or parallel data transfers could further improve performance.
- B) A database system is deployed over a wide area network where communication costs are significantly higher than local processing costs (communication to I/O cost ratio = 20:1). A developer frequently executes the same query from a Java application. (3)
- Apply your understanding of cost-based query optimization and query processing to recommend how this query should be executed for maximum efficiency in the given environment.
- C) A query optimizer aims to reduce total query cost in a distributed environment by minimizing communication between sites. (2)
- Explain how minimizing the number of messages and amount of transmitted data helps reduce communication cost in distributed query processing.
- 5) Imagine a distributed e-commerce system where multiple users can simultaneously book the same flight tickets. Apply a suitable distributed concurrency control mechanism to ensure that only one user successfully books a ticket for a specific flight and seat, and then analyze the trade-offs of your chosen mechanism compared to the other options. (5)
- A)

- B) A distributed database uses Multiversion Timestamp Ordering (MVTO) to manage concurrent transactions. When two transactions attempt to read and write the same data item, the system maintains multiple versions of the item to avoid conflicts. Apply the steps of multiversion timestamp ordering to explain how the database system handles concurrent read and write operations on the same data item. Describe how timestamps are used to decide whether a transaction should read an existing version, create a new version, or be aborted. (3)
- C) Given the wait-for relationships $T1 \rightarrow T2 \rightarrow T3 \rightarrow T4 \rightarrow T1$ and the transaction distribution where T1 and T2 run at site 1, and T3 and T4 run at site 2, apply the concept of a Global Wait-For Graph (GWFG) to determine whether a deadlock exists, and justify your conclusion. (2)

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